

Me trying to clean the data

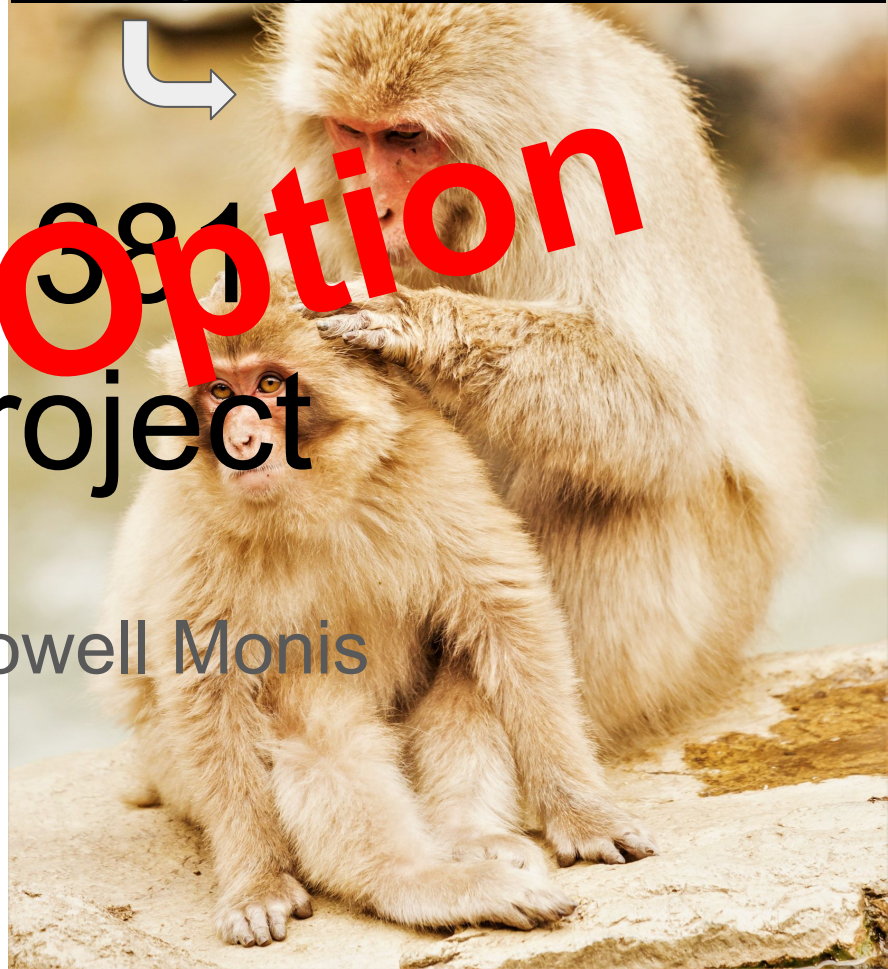


CMSE 381 Final Project

Honors Option

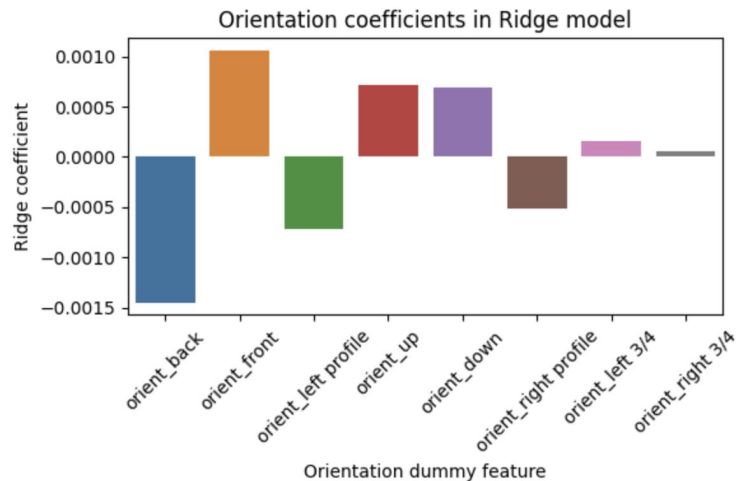
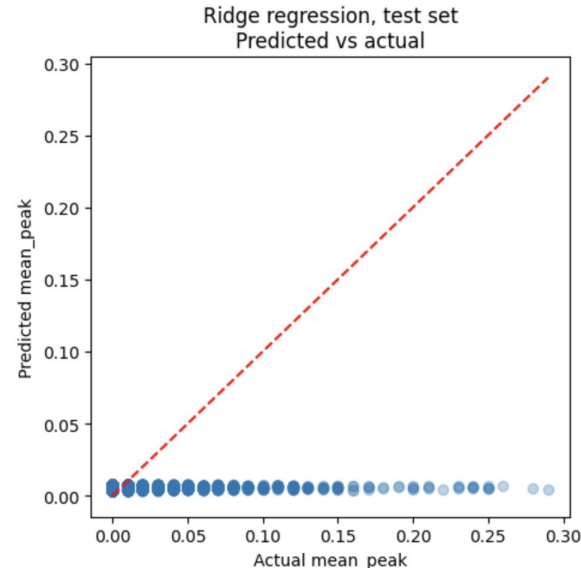
Arkesh Das, Lowell Monis

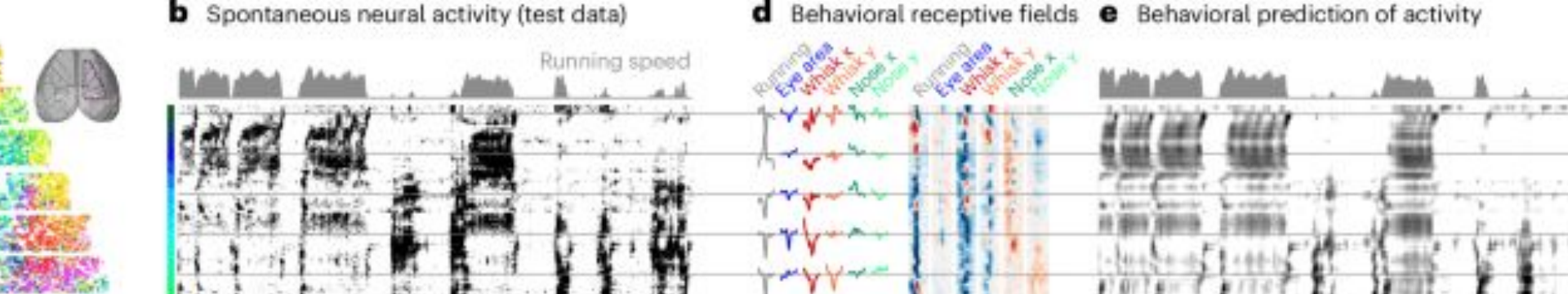
[https://commons.wikimedia.org/wiki/File:Japanese_Snow_Monkey_\(Macaque\)_Mother_Grooms_Her_Young.jpg](https://commons.wikimedia.org/wiki/File:Japanese_Snow_Monkey_(Macaque)_Mother_Grooms_Her_Young.jpg)



Quick Recap

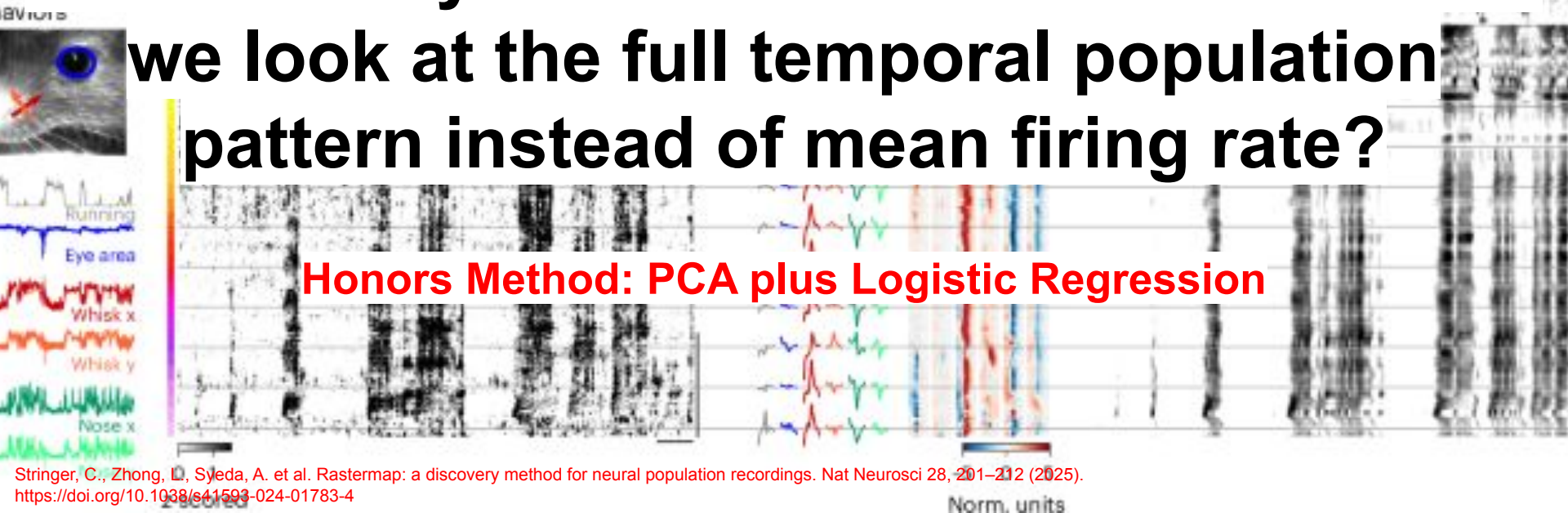
- **RQ1:** How much information about face identity and head orientation is present in AM neural responses?
- One hot encode identity + orientation
- Ridge Regression with cross validation
 - $R^2 < 1$ percent
 - Orientation predicts slightly more than identity
 - Coefficients are extremely small
- AM coding is sparse and nonlinear
- Motivates PCA to analyze full temporal population patterns





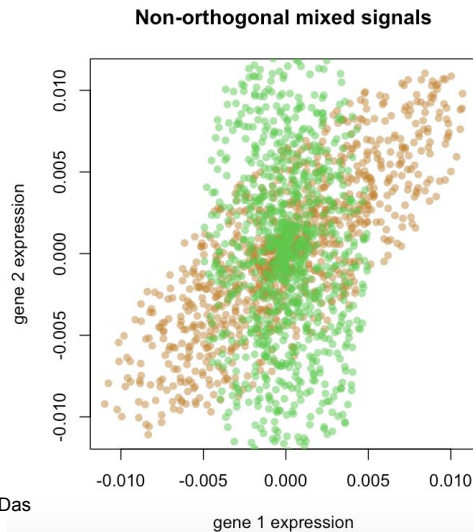
Does identity become more decodable when we look at the full temporal population pattern instead of mean firing rate?

Honors Method: PCA plus Logistic Regression

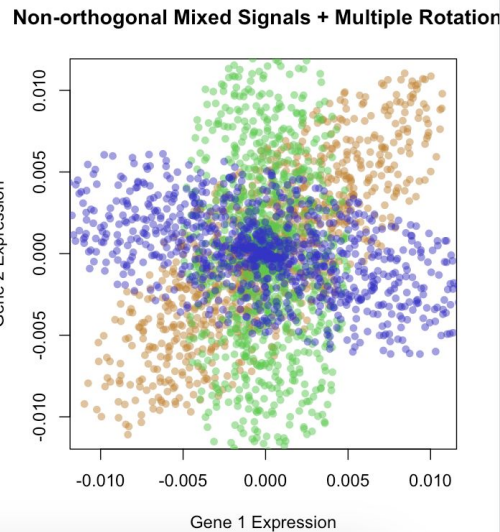


Methods: PCA

- Identity coding in AM is population based
- PCA finds dominant axes of variation across neurons and time
- Logistic regression tests how much identity survives after compression
- Fits the structure of the original AM analyses



Generated in R by Arkesh Das

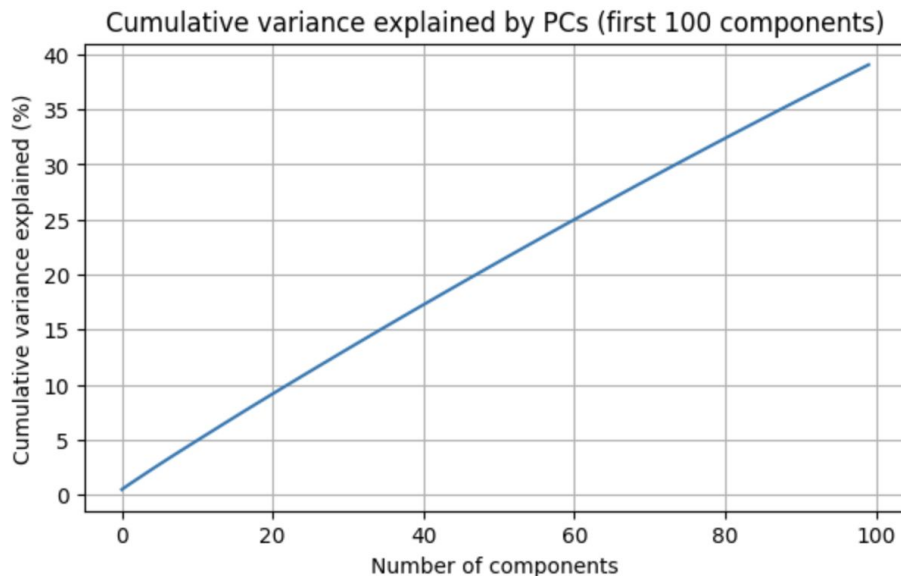


Methods: Data Preprocessing

- Horizontal pseudo population created from vertical single neuron rasters
- Some neurons were recorded for fewer stimulus repetitions, creating NaNs for missing trials
- Restricted analysis to the 281 trials shared across all neurons
- This removed NaNs without inventing fake spikes or zeros

Key Findings from PCA

- First 100 PCs explain about 40 percent
- No clear drop in variance across components
- Identity, if present, is weak relative to other sources of variance



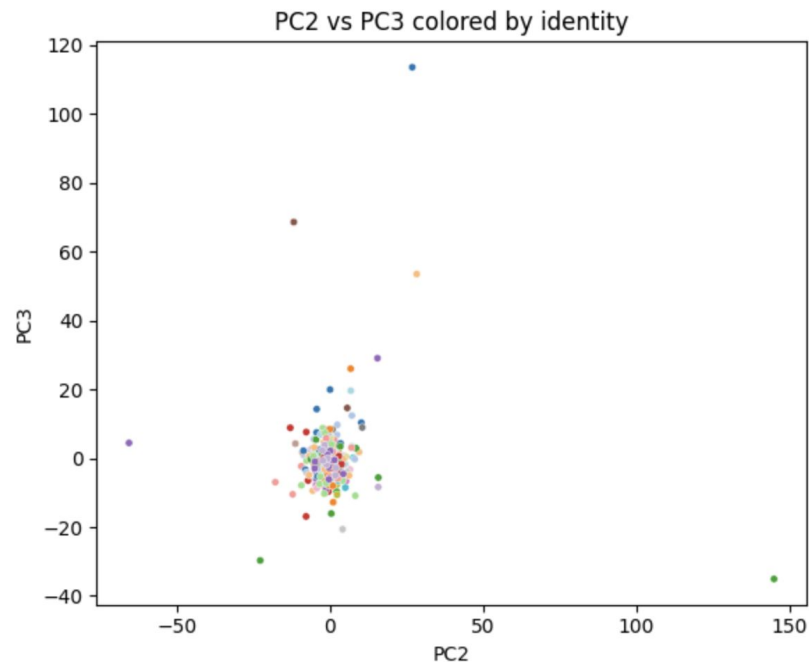
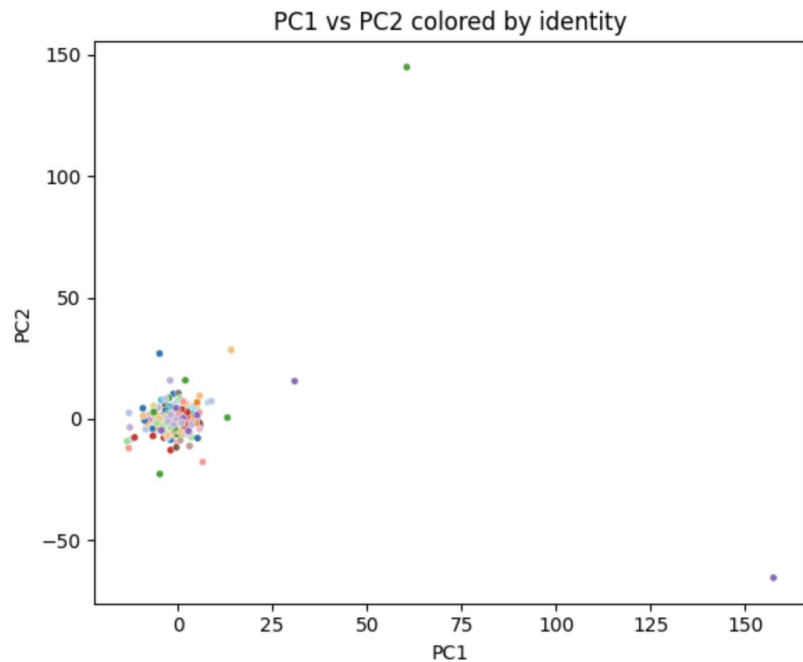
Logistic Regression Accuracy on PCs

Accuracy by number of PCs:

- 5 PCs: about 4.3 percent
- 10 PCs: about 7.5 percent
- 20 PCs: about 5.7 percent
- 50 PCs: about 10.0 percent

Just pure guessing would be 1 out of 25 (4%)

PCA Scatter Plots



Conclusions and Next Steps

- Identity is not visible in mean firing rates
- Identity is not clearly separable in major PCA dimensions
- AM coding appears sparse, distributed, and nonlinear
- Future extensions: nonlinear methods and clustering

Soo Hyun Park *et al.*, Parallel functional subnetworks embedded in the macaque face patch system. *Sci. Adv.* **8**, eabm2054 (2022). DOI:10.1126/sciadv.abm2054

